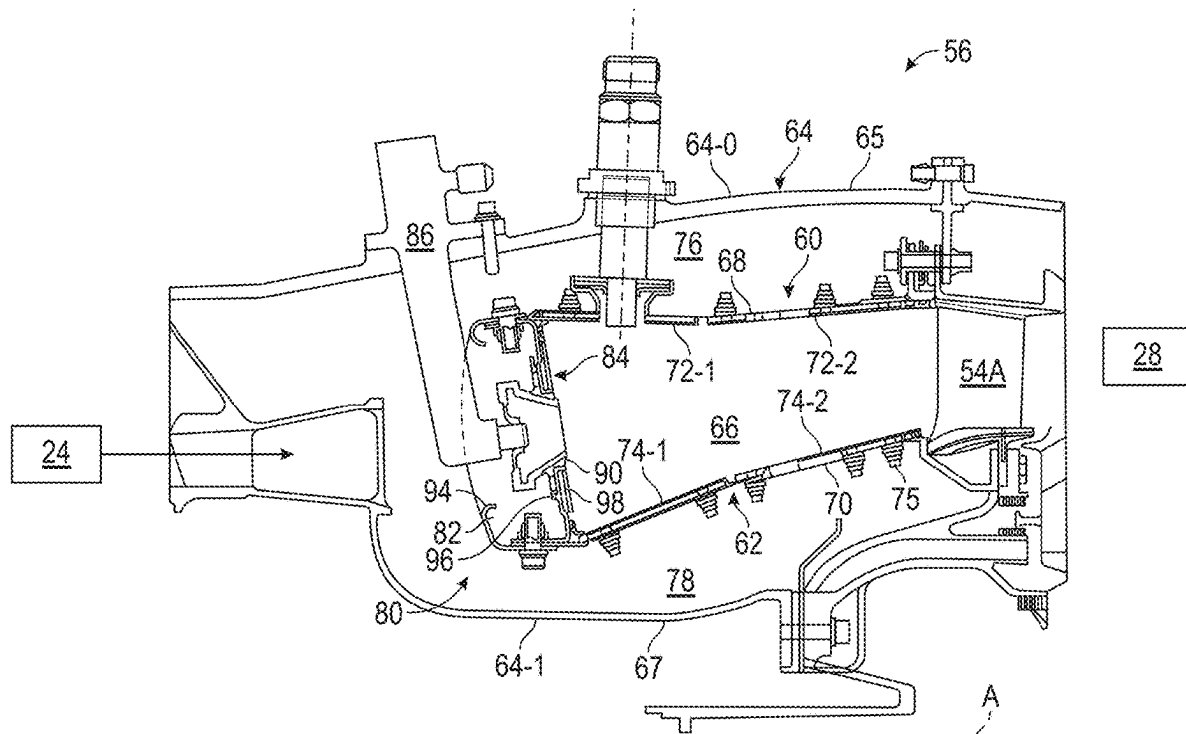




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(19) **United States**(12) **Patent Application Publication**
Snyder et al.(10) **Pub. No.: US 2025/0283599 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **DILUTION PASSAGES FOR COMBUSTOR
OF A GAS TURBINE ENGINE**(52) **U.S. Cl.**
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CT (US)(21) Appl. No.: **18/596,302**(22) Filed: **Mar. 5, 2024****Publication Classification**(51) **Int. Cl.**
F23R 3/00 (2006.01)(57) **ABSTRACT**

A wall assembly for use in a combustor of a gas turbine engine, the wall assembly including: a support shell; a liner panel; and an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage; a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and cooling airflow paths extending from a surface of the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.



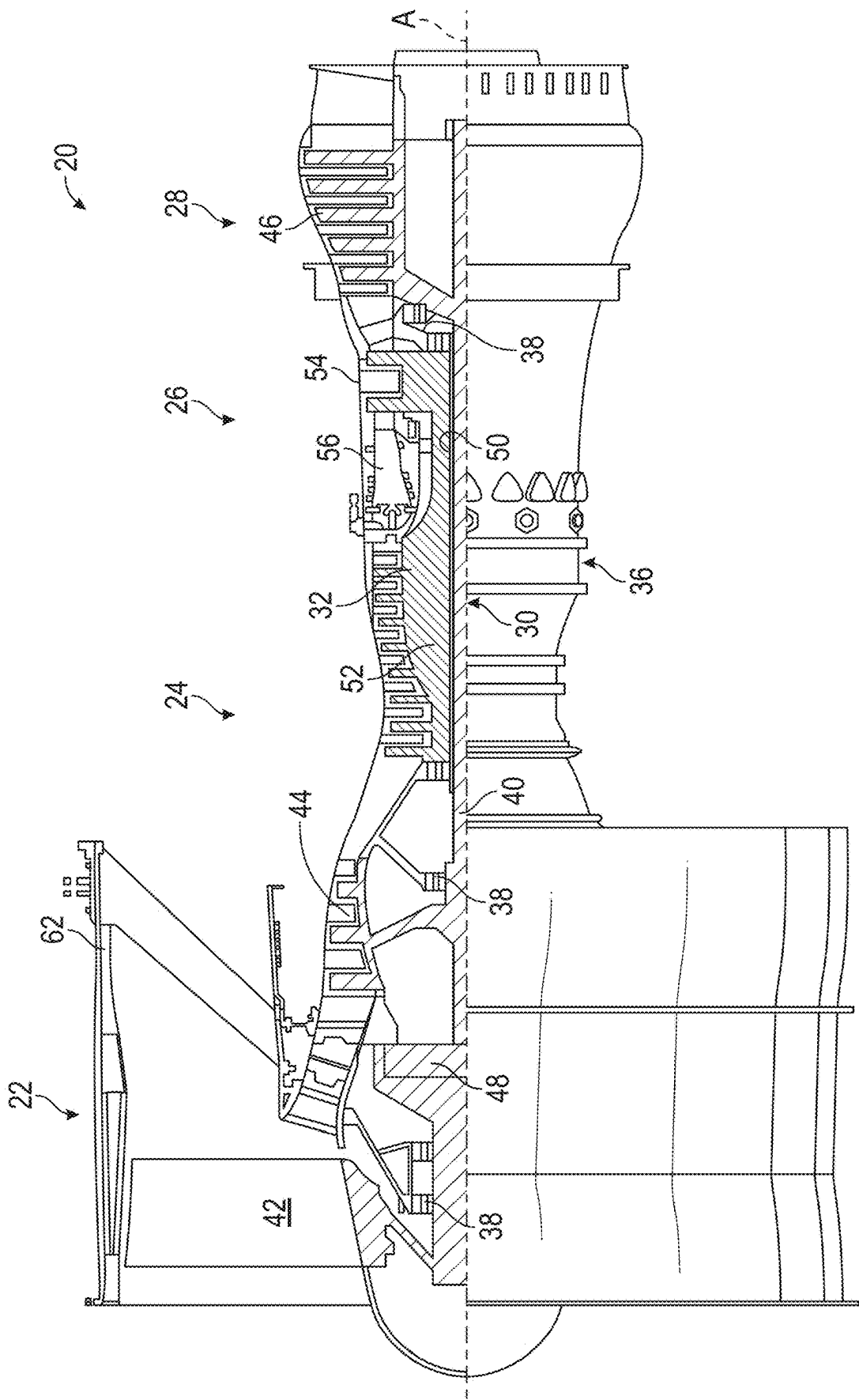


FIG. 1

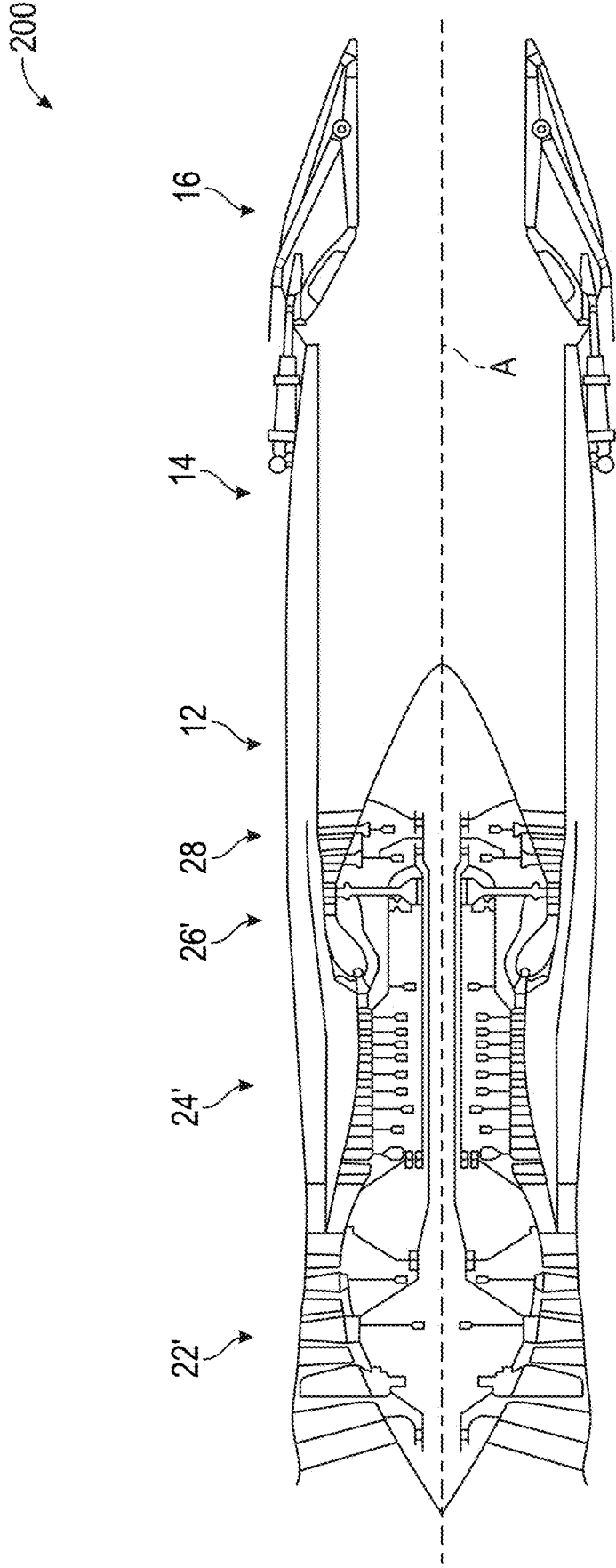


FIG. 2

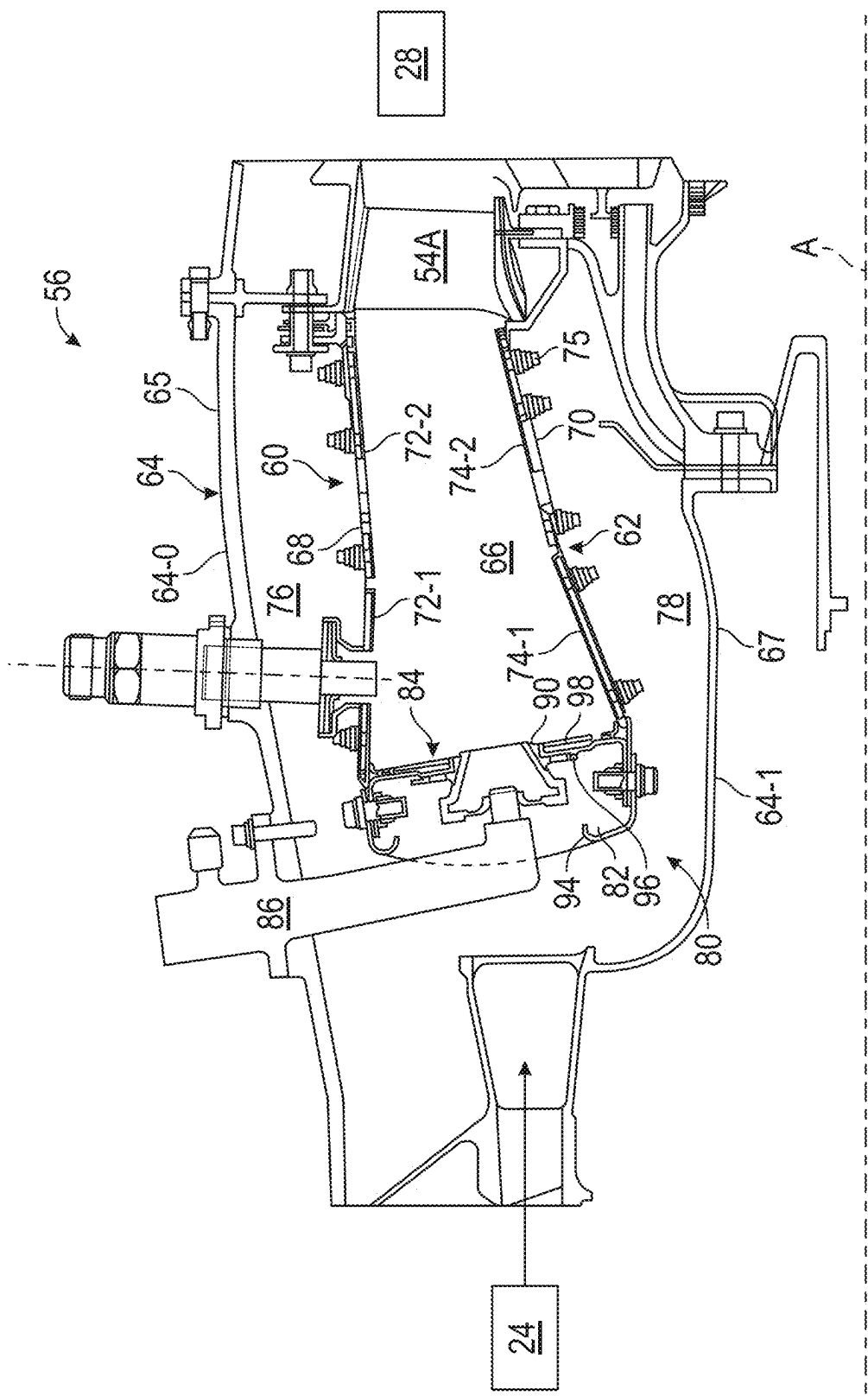


FIG. 3

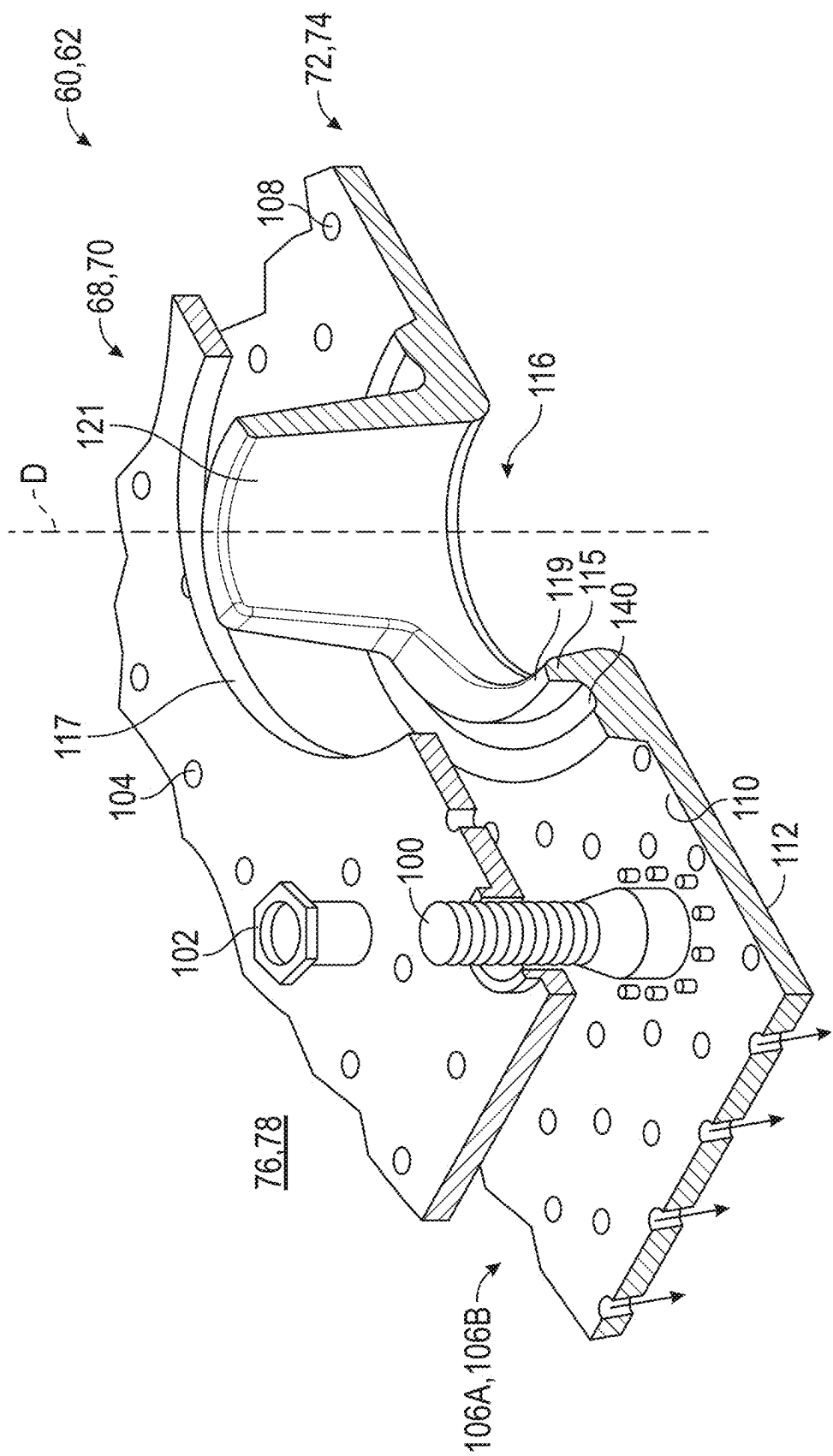


FIG. 4

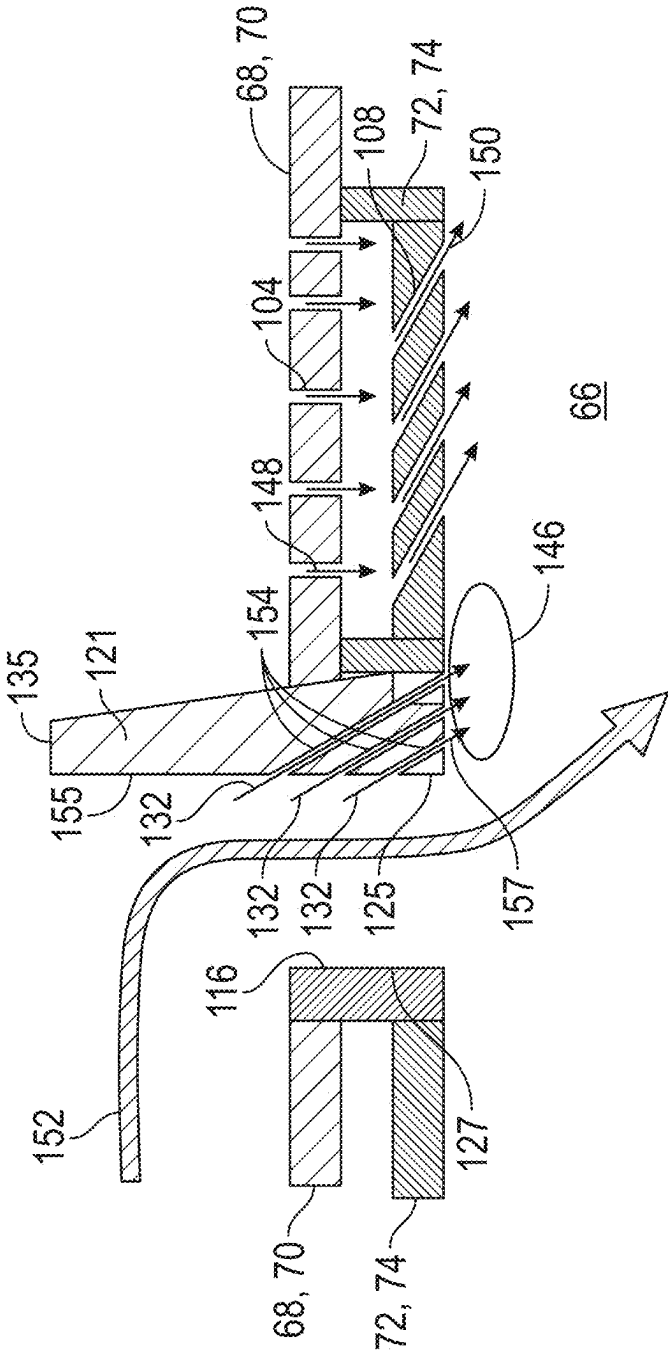


FIG. 7

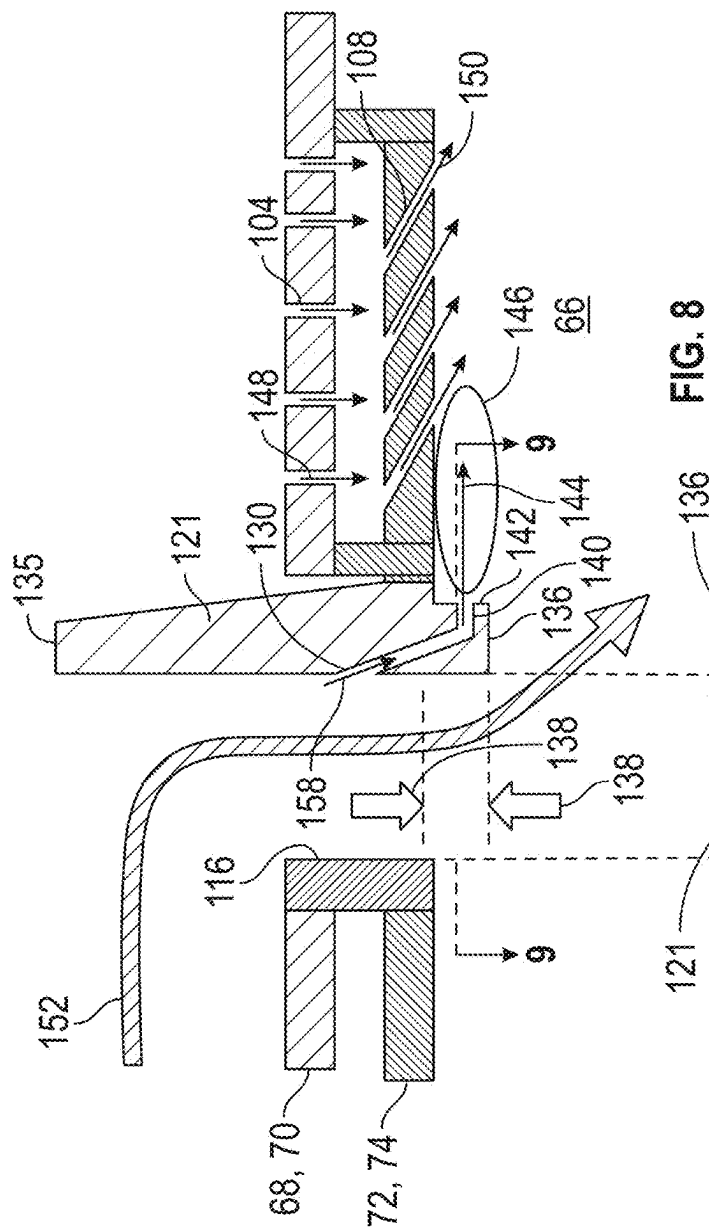


FIG. 8

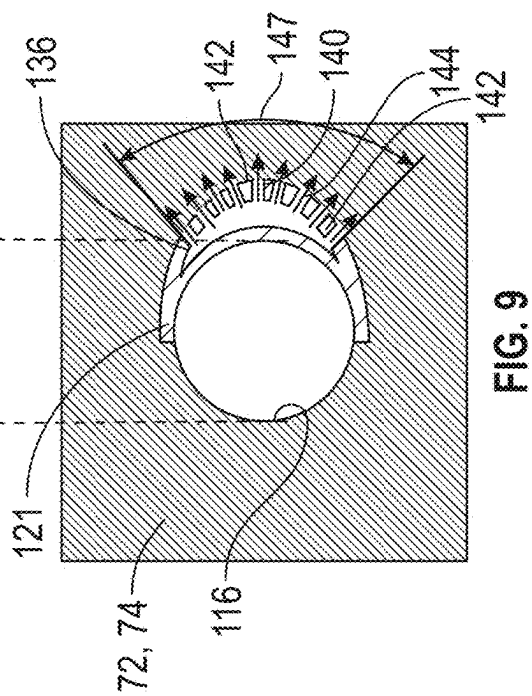


FIG. 9

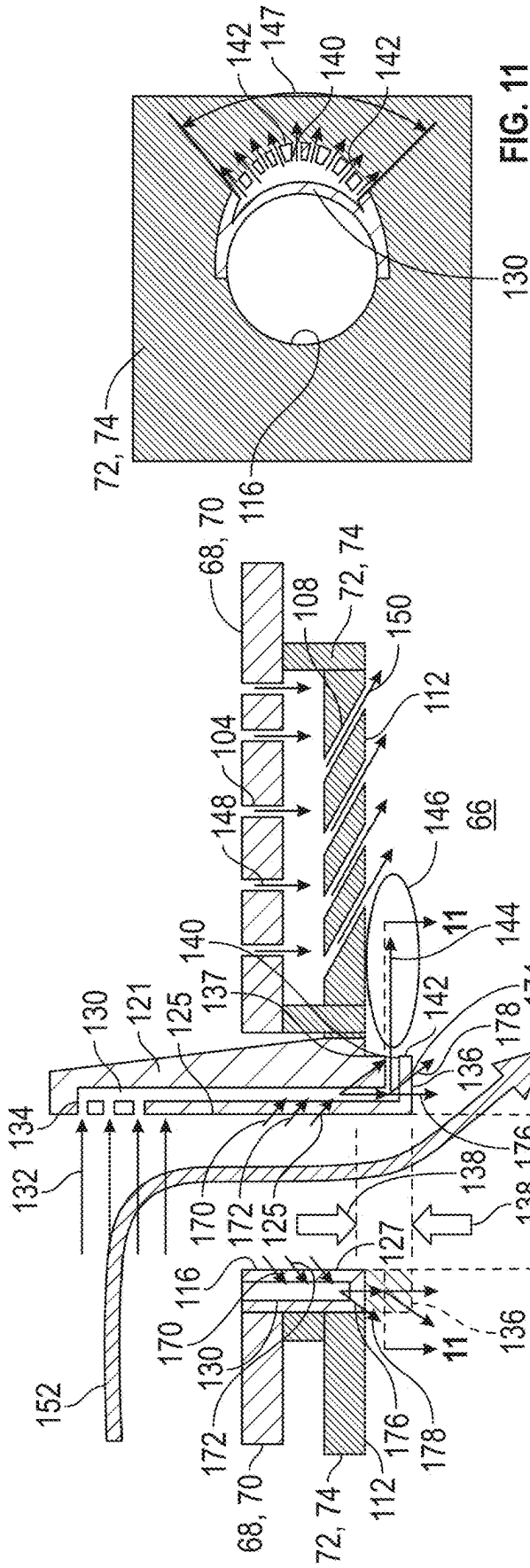


FIG. 10

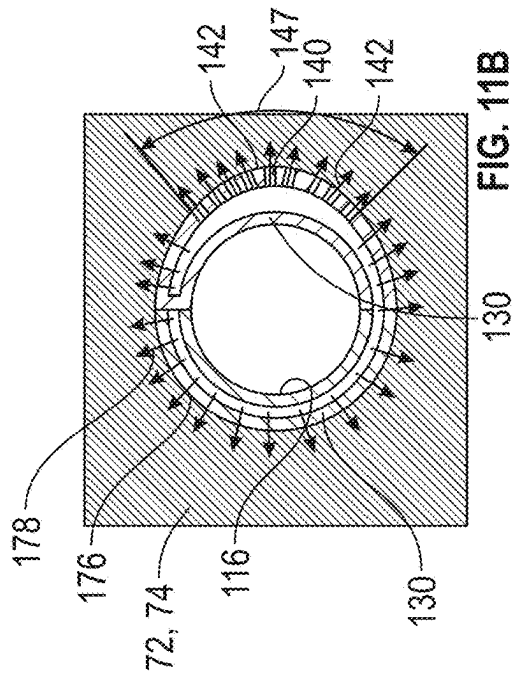


FIG. 11B

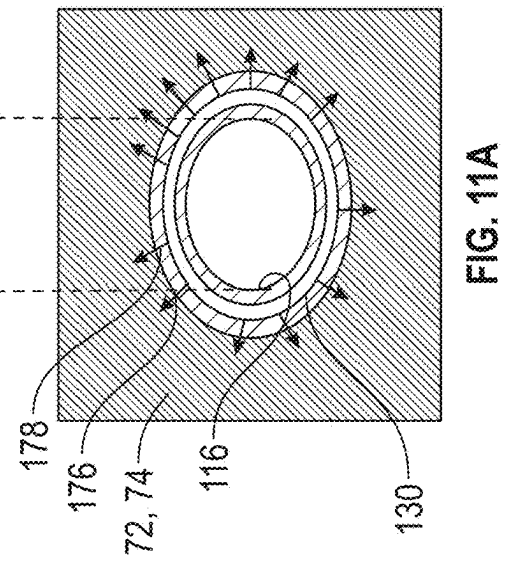


FIG. 11A

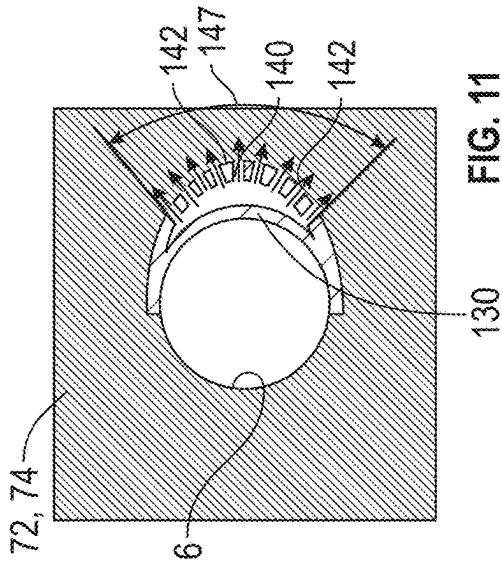


FIG. 11

DILUTION PASSAGES FOR COMBUSTOR OF A GAS TURBINE ENGINE

BACKGROUND

[0001] The present disclosure relates to gas turbine engines and, more particularly, to a combustor section therefor.

[0002] Gas turbine engines typically include a compressor section, a combustor section and a turbine section. During operation, air is pressurized in the compressor section and is mixed with fuel and burned in the combustor section to generate hot combustion gases. The hot combustion gases are communicated through the turbine section, which extracts energy from the hot combustion gases to power the compressor section and other gas turbine engine loads.

[0003] The combustor section typically includes an outer shell lined with heat shields often referred to as floatwall panels which are attached to the outer shell with studs and nuts. In certain arrangements, dilution holes in the floatwall panel communicate with respective dilution holes in the outer shell to direct cooling air for dilution of the combustion gases. In addition to the dilution holes, the outer shell may also have relatively smaller air impingement holes to direct cooling air between the floatwall panels and the outer shell to cool the cold side of the floatwall panels. This cooling air exits effusion holes on the surface of the floatwall panels to form a film on a hot side of the floatwall panels which serves as a barrier against thermal damage.

[0004] One particular region where localized hot spots may arise is around the combustor dilution holes. The dilution holes inject relative lower temperature air into the swirling fuel-rich cross flow for combustion. As the air penetrates into the fuel-rich cross-stream, heat release takes place along the reaction front creating high temperature regions around the dilution holes. A stagnation region along the upstream side of the dilution jets also forms a higher pressure environment such that cross flow momentum deflects the incoming dilution jet. It is the combination of high pressure and the deflection of the incoming jet which is believed to create a high temperature recirculation region along the inner surface of the dilution hole. These high temperatures recirculation regions may affect the life of the turbine of the gas turbine engine as well as the cooling design of the turbine.

[0005] A lower velocity region of flow along the perimeter of the dilution hole may be highly susceptible to inflow of hot combustion gas products. The inflow of these products can occur within a localized ingestion region and may result in a durability concern because a low temperature boundary condition is replaced by high temperature gases.

BRIEF DESCRIPTION

[0006] Disclosed is a wall assembly for use in a combustor of a gas turbine engine, the wall assembly including: a support shell; a liner panel; and an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage; a backstop extending from the top

portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

[0007] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an internal plenum is located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum.

[0008] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a plurality of openings are located in the surface of the backstop above the support shell, the plurality of openings being in fluid communication with the internal plenum.

[0009] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a protrusion extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion.

[0010] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the at least one opening is a plurality of individual openings or a single elongated slot.

[0011] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the protrusion has a curvature that matches a curvature of the dilution passage.

[0012] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of individual openings or the single elongated slot are arranged in an arc.

[0013] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the backstop is arranged to be at least partially located at a trailing edge portion of the dilution passage.

[0014] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a plurality of openings are located in the surface of the backstop, a surface of the annular peripheral wall defining the dilution passage and a bottom portion of the annular peripheral wall, the plurality of opening being in fluid communication with the cooling airflow paths and the cooling airflow paths are angled with respect to the surface of the backstop and the bottom portion of the annular peripheral wall.

[0015] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, some of the plurality of openings located in the bottom portion the annular peripheral wall are further from a trailing edge of the dilution passage than other ones of the plurality of openings located in the bottom portion of the annular peripheral wall, the some of the plurality of openings located in the bottom portion of the annular peripheral wall that are further from the trailing edge of the dilution passage correspond to some of the plurality of openings located in the surface of the backstop or the surface of the annular peripheral wall that are further from the bottom portion of the annular peripheral wall than others of the

plurality of openings located in the surface of the backstop or the surface of the annular peripheral wall.

[0016] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an internal plenum is located in the annular peripheral wall, the cooling airflow paths extending through the internal plenum and at least one opening located in a surface of the annular peripheral wall defining the dilution passage, the at least one opening being in fluid communication with the internal plenum; a protrusion extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion.

[0017] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the at least one opening is located in the trailing edge portion of the protrusion is a plurality of individual openings or a single elongated slot.

[0018] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the protrusion has a curvature that matches a curvature of the dilution passage.

[0019] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of individual openings or the single elongated slot are arranged in an arc.

[0020] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an internal plenum is located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum; and a plurality of openings are located in the surface of the backstop above the support shell, a surface of the annular peripheral wall defining the dilution passage and a bottom portion of the annular peripheral wall, the plurality of openings being in fluid communication with the internal plenum.

[0021] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an internal plenum is located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum; a protrusion extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion; a plurality of openings located in the surface of the backstop above the support shell, and a surface of the annular peripheral wall defining the dilution passage, the plurality of openings and the at least one opening located in a trailing edge portion of the protrusion being in fluid communication with the internal plenum.

[0022] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the protrusion completely encircles the dilution passage and a bottom portion of the protrusion has openings in fluid communication with the internal plenum.

[0023] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the internal plenum extends completely around an interior of the annular peripheral wall.

[0024] Also disclosed is a gas turbine engine, including: a compressor section; a turbine section; and a combustor section, the combustor section including a wall assembly for use in a combustor of the combustor section, the wall assembly including: a support shell; a liner panel; and an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage; a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

[0025] Also disclosed is a method of guiding airflow into a combustor of a gas turbine engine, the method including: locating a portion of an annular peripheral wall in an opening of a support shell of the combustor, the annular peripheral wall extending from a liner panel secured to the support shell, the annular peripheral wall defining a dilution passage; and directing airflow into the dilution passage via a backstop extending from a top portion of the annular peripheral wall, the top portion defining a portion of a periphery of the dilution passage; and directing airflow into cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

[0027] FIG. 1 is a schematic, partial cross-sectional view of a gas turbine engine in accordance with this disclosure;

[0028] FIG. 2 is schematic cross-section of another example gas turbine engine;

[0029] FIG. 3 is an expanded longitudinal schematic sectional view of a combustor section that may be used with the example gas turbine engine architectures shown in FIGS. 1 and 2;

[0030] FIG. 4 is an exploded view of a wall assembly with a dilution passage in accordance with the present disclosure;

[0031] FIG. 5 is a cross-sectional view of a dilution passage in accordance with an embodiment of the present disclosure;

[0032] FIG. 6 is a view along lines 6-6 of FIG. 5;

[0033] FIG. 7 is a cross-sectional view of a dilution passage in accordance with an alternative embodiment of the present disclosure;

[0034] FIG. 8 is a cross-sectional view of a dilution passage in accordance with an alternative embodiment of the present disclosure;

[0035] FIG. 9 is a view along lines 9-9 of FIG. 8;

[0036] FIG. 10 is a cross-sectional view of a dilution passage in accordance with another embodiment of the present disclosure;

[0037] FIG. 11 is a view along lines 11-11 of FIG. 10 in accordance with another embodiment of the present disclosure;

[0038] FIG. 11A is a view along lines 11-11 of FIG. 10 in accordance with another embodiment of the present disclosure; and

[0039] FIG. 11B is a view along lines 11-11 of FIG. 10 in accordance with another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0040] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the FIGS.

[0041] FIG. 1 schematically illustrates a gas turbine engine 20. The gas turbine engine 20 is disclosed herein as a two-spool turbo fan that generally incorporates a fan section 22, a compressor section 24, a combustor section 26 and a turbine section 28. Referring to FIG. 2, alternative engine architectures 200 might include additional sections 12, 14, and 16 in addition to the fan section 22', compressor section 24', combustor section 26' and turbine section 28' among other systems or features. Referring again to FIG. 1, the fan section 22 drives air along a bypass flowpath while the compressor section 24 drives air along a core flowpath for compression and communication into the combustor section 26 then expansion through the turbine section 28. Although depicted as a turbofan in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with turbofans as the teachings may be applied to other types of turbine engines such as a turbojets, turboshafts, and three-spool (plus fan) turbofans wherein an intermediate spool includes an intermediate pressure compressor ("IPC") between a low pressure compressor ("LPC") and a high pressure compressor ("HPC"), and an intermediate pressure turbine ("IPT") between a high pressure turbine ("HPT") and a low pressure turbine ("LPT").

[0042] The engine 20 generally includes a low spool 30 and a high spool 32 mounted for rotation about an engine central longitudinal axis A relative to an engine static structure 36 via several bearing structures 38. The low spool 30 generally includes an inner shaft 40 that interconnects a fan 42, a low pressure compressor ("LPC") 44 and a low pressure turbine ("LPT") 46. The inner shaft 40 may drive the fan 42 directly, or through a geared architecture 48 as illustrated in FIG. 1 to drive the fan 42 at a lower speed than the low spool 30. An exemplary reduction transmission is an epicyclic transmission, namely a planetary or star gear system.

[0043] The high spool 32 includes an outer shaft 50 that interconnects a high pressure compressor ("HPC") 52 and a high pressure turbine ("HPT") 54. A combustor 56 is arranged between the high pressure compressor 52 and the high pressure turbine 54. The inner shaft 40 and the outer shaft 50 are concentric and rotate about the engine central longitudinal axis A which is collinear with their longitudinal axes.

[0044] Core airflow is compressed by the low pressure compressor 44 then the high pressure compressor 52, mixed with the fuel and burned in the combustor 56, then expanded over the high pressure turbine 54 and the low pressure turbine 46. The low pressure turbine 46 and the high

pressure turbine 54 rotationally drive the respective low spool 30 and the high spool 32 in response to the expansion. The main engine shafts 40, 50 are supported at a plurality of points by the bearing structures 38 within the static structure 36. It should be understood that various bearing structures 38 at various locations may alternatively or additionally be provided.

[0045] With reference to FIG. 3, the combustor section 26, 26' generally includes the combustor 56 with an outer combustor wall assembly 60, an inner combustor wall assembly 62 and a diffuser case module 64 therearound. The outer combustor wall assembly 60 and the inner combustor wall assembly 62 are spaced apart such that an annular combustion chamber 66 is defined therebetween.

[0046] The outer combustor wall assembly 60 is spaced radially inward from an outer diffuser case 64-O of the diffuser case module 64 to define an outer annular plenum 76. The inner combustor wall assembly 62 is spaced radially outward from an inner diffuser case 64-I of the diffuser case module 64 to define an inner annular plenum 78. It should be understood that although a particular combustor is illustrated, other combustor types with various combustor liner arrangements will also benefit therefrom. It should be further understood that the disclosed cooling flow paths are but an illustrated embodiment and should not be limited only thereto.

[0047] The combustor wall assemblies 60, 62 contain the combustion products for direction toward the turbine section 28. Each combustor wall assembly 60, 62 generally includes a respective support shell 68, 70 which supports one or more liner panels 72, 74 mounted thereto. Each of the liner panels 72, 74 may be generally rectilinear and manufactured of, for example, a nickel based super alloy, ceramic or other temperature resistant material and are arranged to form a liner array. In the liner array, a multiple of forward liner panels 72A and a multiple of aft liner panels 72B are circumferentially staggered to line the outer shell 68. A multiple of forward liner panels 74A and a multiple of aft liner panels 74B are circumferentially staggered to also line the inner shell 70.

[0048] The combustor 56 further includes a forward assembly 80 immediately downstream of the compressor section 24 to receive compressed airflow therefrom. The forward assembly 80 generally includes an annular hood 82, a bulkhead assembly 84, and a multiple of swirlers 90 (one shown). Each of the swirlers 90 is circumferentially aligned with one of a multiple of fuel nozzles 86 (one shown) and the respective hood ports 94 to project through the bulkhead assembly 84. The bulkhead assembly 84 includes a bulkhead support shell 96 secured to the combustor walls 60, 62, and a multiple of circumferentially distributed bulkhead liner panels 98 secured to the bulkhead support shell 96 around each respective swirler opening 92. The bulkhead support shell 96 is generally annular and the multiple of circumferentially distributed bulkhead liner panels 98 are segmented, typically one to each fuel nozzle 86 and swirler 90.

[0049] The annular hood 82 extends radially between, and is secured to, the forwardmost ends of the combustor wall assemblies 60, 62. The annular hood 82 includes the multiple of circumferentially distributed hood ports 94 that receive one of the respective multiple of fuel nozzles 86 and facilitates the direction of compressed air into the forward end of the combustion chamber 66 through a respective one of the swirler openings 92. Each fuel nozzle 86 may be

secured to the diffuser case module 64 and project through one of the hood ports 94 into the respective swirler 90.

[0050] The forward assembly 80 introduces core combustion air into the forward section of the combustion chamber 66 while the remainder enters the outer annular plenum 76 and the inner annular plenum 78. The multiple of fuel nozzles 86 and adjacent structure generate a blended fuel-air mixture that supports stable combustion in the combustion chamber 66.

[0051] Opposite the forward assembly 80, the outer and the inner support shells 68, 70 are mounted adjacent to a first row of nozzle guide vanes (NGVs) 54A in the high pressure turbine 54. The nozzle guide vanes 54A are static engine components which direct core airflow combustion gases onto the turbine blades of the first turbine rotor in the turbine section 28 to facilitate the conversion of pressure energy into kinetic energy. The core airflow combustion gases are also accelerated by the nozzle guide vanes 54A because of their convergent shape and the gases are typically given a “spin” or a “swirl” in the direction of turbine rotor rotation. The turbine rotor blades absorb this energy to drive the turbine rotor at high speed.

[0052] With reference to FIG. 4, a plurality of studs 100 extend from the liner panels 72, 74 so as to permit the liner panels 72, 74 to be mounted to their respective support shells 68, 70 with fasteners 102 such as nuts. That is, the studs 100 project rigidly from the liner panels 72, 74 and through the respective support shells 68, 70 to receive the fasteners 102 at a threaded distal end section thereof.

[0053] A multiple of cooling impingement passages 104 penetrate through the support shells 68, 70 to allow air from the respective annular plenums 76, 78 to enter cavities 106A, 106B formed in the combustor wall assemblies 60, 62 between the respective support shells 68, 70 and liner panels 72, 74. The cooling impingement passages 104 are generally normal to the surface of the liner panels 72, 74. The air in the cavities 106A, 106B provides cold side impingement cooling of the liner panels 72, 74. As used herein, the term impingement cooling generally implies heat removal from a part via an impinging gas jet directed at a part.

[0054] A multiple of effusion passages 108 penetrate through each of the liner panels 72, 74. The geometry of the passages (e.g., diameter, shape, density, surface angle, incidence angle, etc.) as well as the location of the passages with respect to the high temperature main flow also contributes to effusion film cooling. The combination of impingement passages 104 and effusion passages 108 may be referred to as an Impingement Film Floatwall (IFF) assembly.

[0055] The effusion passages 108 allow the air to pass from the cavities 106A, 106B defined in part by a cold side 110 of the liner panels 72, 74 to a hot side 112 of the liner panels 72, 74 and thereby facilitate the formation of thin, cool, insulating blanket or film of cooling air along the hot side 112. The effusion passages 108 are generally more numerous than the impingement passages 104 to promote the development of film cooling along the hot side 112 to sheath the liner panels 72, 74. Film cooling as defined herein is the introduction of a relatively cooler air at one or more discrete locations along a surface exposed to a high temperature environment to protect that surface in the region of the air injection as well as downstream thereof.

[0056] A plurality of dilution passages 116 penetrate through both the respective support shells 68, 70 and liner panels 72, 74 along a common axis D. For example only, in

a Rich-Quench-Lean (R-Q-L) type combustor, the dilution passages 116 are located downstream of the forward assembly 80 to quench the hot combustion gases within the combustion chamber 66 by direct supply of cooling air from the respective annular plenums 76, 78. In one non-limiting embodiment, the dilution passages 116 are formed by an annular grommet or annular peripheral wall 115 extending upwardly from the cold side of the liner panels 72, 74. The annular peripheral wall 115 being received within an opening 117 in shells 68, 70 when the liner panels 72, 74 and the shells 68, 70 are assembled together. The annular peripheral wall 115 terminates at a top portion 119 that defines a first periphery of the dilution opening 116 and a backstop or extended wall portion 121 extends upwardly from the top portion 119 of the annular peripheral wall 115. In other words, top portion 119 forms a portion of a periphery of the dilution passage and the backstop 121 forms another portion or the remaining portion of a periphery of the dilution passage 116. In one embodiment, the annular peripheral wall 115, backstop 121 and liner panels 72, 74 are integrally formed as a single unitary structure or in other words, they are formed as a single piece.

[0057] With reference to FIG. 5 and as mentioned above, at least one of the multiple of dilution passages 116 includes a backstop 121 extending upwardly and away from the liner panels 72, 74 proximate to a peripheral portion of the dilution passage 116. The backstop 121 partially circumscribes each or a subset of the dilution passages 116. As used herein a subset of dilution passages may also be referred to as a sector. In one non-limiting embodiment, the backstops 121 are symmetric. However, alternate embodiments could incorporate intentional asymmetries. For example, the backstop 121 could be rotated about the dilution passage centerline D. As illustrated in at least FIG. 4, the backstop 121 may have a curvature that matches a curvature of the dilution opening 116.

[0058] As illustrated and in one embodiment, the backstop 121 is arranged to be at least partially located at a trailing edge portion 125 of the dilution passage 116. The trailing edge portion 125 of the dilution passage 116 is located opposite a leading edge portion 127 of the dilution passage 116. As used herein, the “leading edge” refers to a portion of the dilution passage that faces the incoming airflow and the “trailing edge” refers to a portion of the dilution passage downstream from the “leading edge”. In other words, the leading edge encounters the incoming airflow prior to the trailing edge. In addition, and as used herein, “top” and “bottom” refer to heights or distances from or radially from the engine central longitudinal axis A. The top of an item is radially further from the engine central longitudinal axis A than a bottom of the same item.

[0059] As illustrated in FIG. 5, the backstop 121 as well as the annular peripheral wall 115 has an interior plenum 130. The interior plenum 130 is in fluid communication with a cooling airflow (illustrated by arrows 132) via a plurality of openings 134. The plurality of openings or at least a majority of the plurality of openings 134 are located in the backstop 121 above shells 68, 70 so that they are positioned to capture cooling airflow flowing in the direction of arrows 132, which may be located closer to a top 135 of the backstop 121. This airflow above shells 68, 70 and closer to a top 135 of the backstop 121 may be referred to as a velocity head of cooling airflow.

[0060] The annular peripheral wall 115 has a protrusion 136 that extends from a bottom portion 137 of the annular peripheral wall 115 past the hot side 112 of the liner panels 72, 74. The bottom portion 137 defines at least a portion of a second periphery of the dilution opening 116 and the protrusion 136 forms another portion or the remaining portion of a periphery of the second periphery of the dilution passage 116.

[0061] The distance the protrusion 136 extends past the hot side 112 of the liner panels 72, 74 is illustrated by arrows 138. The protrusion 136 like the backstop 121 partially circumscribes each or a subset of the dilution passages 116. As illustrated, the protrusion 136 like the backstop 121 may have a curvature that matches a curvature of the dilution opening 116. The interior plenum 130 extends into the protrusion and is in fluid communication with the combustion chamber 66 via a plurality of or an array of individual openings 140 located in a trailing edge portion 142 of protrusion 136. In an alternative embodiment, the plurality of individual openings 140 may be replaced by a single elongated slot or opening 140.

[0062] As such, the internal plenum 130 via openings 134 and 140 provides cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall 115 to provide cooling air to a surface 112 of the liner panel 72, 74. The cooling air flow (illustrated by arrows 132) will enter the interior plenum 130 from the backstop 121 and then exit the trailing edge portion 142 of the protrusion 136 via openings 140. The exiting airflow is illustrated by arrows 144. This cooling air will provide surface film cooling to the hot side 112 of the liner panels 72, 74. This surface film cooling is illustrated by area 146. This surface film cooling is downstream of the protrusion 136 and is particularly useful in cooling the hot side 112 of the liner panels 72, 74 downstream of the protrusion 136.

[0063] As illustrated in FIG. 6, the openings or single elongated slot or opening 140 are arranged in an arc 147 in order to provide the surface film cooling illustrated by area 146. This area 146 is located upstream of the surface cooling provided by openings 104 and 108, which is illustrated by arrows 148, 150. As illustrated in FIG. 5, the cooling openings 108 are angled in a downstream orientation to provide cooling air in the direction of arrows 150. This area 146 helps to reduce hot spots adjacent to dilution passages 116 which are hard to cool as they are directly behind the also permits utilization of the relatively limited cooling air elsewhere in the combustor allowing for the more efficient use of combustor airflow. There is also dilution air or cooling air that is jetted into dilution passage 116 that does not enter internal plenum 130. This air is illustrated by arrow 152.

[0064] Referring now to FIG. 7, an alternative embodiment of the present disclosure is illustrated. In this embodiment, there is no protrusion 136 or internal plenum 130. However and in order to provide film cooling at the trailing edge 125 of opening 116, an array or a plurality of cooling paths 154 extending through the backstop 121 as well as the annular peripheral wall 115 defining opening 116. Each of these cooling paths 154 has an inlet opening 155 and an exit opening 157. Each of the inlet openings 155 are arranged at various heights from the trailing edge 125 of a bottom of the opening 116. Each of these inlet openings 155 correspond to an exit opening 157 and each of the cooling paths 154 are inclined with respect to the backstop 121 and the annular peripheral wall such that openings 155 further from the

bottom of the annular peripheral wall 115 have a corresponding exit opening 157 further from the trailing edge 125 of the opening 116 at the bottom of the annular peripheral wall 115. In other words, the higher inclination cooling paths 154 purge cooling air further aft of the bottom of the opening 116 as compared to cooling paths 154 at lower heights in the backstop 121 and/or annular peripheral wall 115. In other words, the corresponding inlet openings 155 located closest to the hot side 112 of the liner panels 72, 74 communicate with exit openings 157 furthest from the trailing edge 125 of a periphery of the bottom of the opening 116 defined by the annular peripheral wall 115. In one non-limiting embodiment, the exit openings 157 may have shaped outlets to diffuse the cooling air at the hot side 112 of the liner panels 72, 74 as well as prevent penetration of hot air back into openings 157 from the hot side 112 of the liner panels 72, 74. As such, openings 157 provide surface film cooling illustrated by area 146 in hard to cool regions behind the dilution hole 116 wake created by the cooling air provided in the direction of arrow 152.

[0065] Referring now to FIGS. 8 and 9, another alternative embodiment of the present disclosure is illustrated. This embodiment is similar to the embodiment illustrated in FIGS. 5 and 6 which includes protrusion 136 however, the internal plenum 130 is shorter in overall height than the internal plenum 130 of the embodiment of FIGS. 5 and 6. In addition, the internal plenum 130 is angled with respect to the interior surface of the annular peripheral wall 115 defining the peripheral surface of the diffusion opening 116. In this embodiment, an inlet or inlets 158 are located in the trailing edge 125 of the annular peripheral wall 115 as opposed to the backstop 121 and are located below the support shells 68, 70 but above the liner panels 72, 74. As such, the inlet or inlets 158 are within the quench hole passage 116 defined by the annular peripheral wall 115. As in the embodiment illustrated in FIGS. 5 and 6, the inlet or inlets 158 are also in fluid communication with the combustion chamber 66 via a plurality of or an array of individual openings 140 located in a trailing edge portion 142 of protrusion 136. In an alternative embodiment, the plurality of individual openings 140 may be replaced by a single elongated slot or opening 140.

[0066] As such, the cooling air flow (illustrated by arrows 132) will enter the interior plenum 130 and then exit the trailing edge portion 142 of the protrusion 136 via openings 140. The exiting airflow is illustrated by arrows 144. This cooling air will provide surface film cooling to the hot side 112 of the liner panels 72, 74. This surface film cooling is illustrated by area 146. This surface film cooling is downstream of the protrusion 136 and is particularly useful in cooling the hot side 112 of the liner panels 72, 74 downstream of the protrusion 136.

[0067] Referring now to FIGS. 10-11B, another alternative embodiment of the present disclosure is illustrated. Here the backstop 121 and the annular peripheral wall 115 has an interior plenum 130. The interior plenum 130 is in fluid communication with a cooling airflow (illustrated by arrows 132) via a plurality of openings 134. The plurality of openings or at least a majority of the plurality of openings 134 are located in a surface of the backstop 121 above shells 68, 70 so that they are positioned to capture cooling airflow flowing in the direction of arrows 132, which may be located closer to a top 135 of the backstop 121. This airflow above

shells 68, 70 and closer to a top 135 of the backstop 121 may be referred to as a velocity head of cooling airflow.

[0068] The annular peripheral wall 115 has a protrusion 136 that extends from a bottom portion 137 of the annular peripheral wall past the hot side 112 of the liner panels 72, 74. The distance the protrusion 136 extends past the hot side 112 of the liner panels 72, 74 is illustrated by arrows 138. The protrusion 136 like the backstop 121 partially circumscribes each or a subset of the dilution passages 116. As illustrated, the protrusion 136 like the backstop 121 may have a curvature that matches a curvature of the dilution opening 116. The interior plenum 130 extends through the backstop 121 and the annular peripheral wall 115 and is also in fluid communication with the combustion chamber 66 via a plurality of or an array of individual openings 140 located in a trailing edge portion 142 of protrusion 136. In an alternative embodiment, the plurality of individual openings 140 may be replaced by a single elongated slot or opening 140.

[0069] In addition, the interior plenum 130 is also in fluid communication with a plurality or array of openings 170 located in a surface the annular peripheral wall 115 arranged at a trailing edge portion 125 of the dilution passage 116. The openings 170 are located in the opening 116 such that cooling air in the dilution opening 116 is fed into the interior plenum 130 (illustrated by arrows 172) via cooling openings 170.

[0070] As such, the cooling air flow (illustrated by arrows 132 and 172) will enter the interior plenum 130 and then exit the trailing edge portion 142 of the protrusion 136 via openings 140. The exiting airflow is illustrated by arrows 144. This cooling air will provide surface film cooling to the hot side 112 of the liner panels 72, 74. This surface film cooling is illustrated by area 146. This surface film cooling is downstream of the protrusion 136 and is particularly useful in cooling the hot side 112 of the liner panels 72, 74 downstream of the protrusion 136.

[0071] In an alternative embodiment or configuration, a bottom surface 174 of the protrusion 136 facing the combustion chamber 66 is also provided with a plurality of openings 176 in fluid communication with the interior plenum 130 such that cooling air (illustrated by arrows 178) may be provided to surface 174. Openings 176 may comprises vanes, slots, or shaped holes. The openings 176 may be angled or normal to surface 174.

[0072] As illustrated in FIG. 11, the openings or single elongated slot or opening 140 are arranged in an arc 147 in order to provide the surface film cooling illustrated by area 146. This area 146 is located upstream of the surface cooling provided by openings 104 and 108, which is illustrated by arrows 148, 150. As illustrated in FIG. 10, the cooling openings 108 are angled in a downstream orientation to provide cooling air in the direction of arrows 150. This area 146 helps to reduce hot spots adjacent to dilution passages 116 which are hard to cool as they are directly behind the also permits utilization of the relatively limited cooling air elsewhere in the combustor allowing for the more efficient engine operation. There is also dilution air or cooling air that is jetted into dilution passage 116 that does not enter internal plenum 130. This air is illustrated by arrow 152.

[0073] In yet another alternative embodiment and referring now to FIGS. 10 and 11A, the internal plenum 130 extends completely around opening 116 while being located in the annular peripheral wall 115. This is in contrast to some

of the preceding embodiments where the internal plenum 130 was only located inside some of the annular peripheral wall 115 that defines opening 116. In addition, the entire inner surface of opening 116 including the leading edge portion 127 of the opening 116 is provided with openings 170 so that cooling air can enter plenum 130 about the entire periphery of the opening 116 in the direction of arrows 172. In addition and similar to the previous embodiment, a bottom of the annular peripheral wall 115 as well as the bottom surface 174 of the protrusion 136 facing the combustion chamber 66 is also provided with a plurality of openings 176 in fluid communication with the interior plenum 130 such that cooling air (illustrated by arrows 178) may be provided to surface 112.

[0074] In yet another alternative embodiment and referring now to FIGS. 10 and 11B, the protrusion 136 does not only extend partially from the bottom of the annular peripheral wall 115 but is located around the entire periphery of opening 116 or the bottom of the annular peripheral wall 115 essentially extending the entire annular peripheral wall 115 past surface 112 of the hot side 112 of the liner panels 72, 74. See the dashed lines in FIG. 10.

[0075] In this embodiment and similar to the previous embodiment, the internal plenum 130 extends completely around opening 116. In addition, the entire inner surface of opening 116 including the leading edge portion 127 of the opening 116 is provided with openings 170 so that cooling air can entire plenum 130 about the entire periphery of the opening 116 in the direction of arrows 172. In this embodiment and since the protrusion 136 extends about the entire periphery of opening 116, the bottom surface 174 about the entire periphery of the opening 116 facing the combustion chamber 66 is also provided with a plurality of openings 176 in fluid communication with the interior plenum 130 such that cooling air (illustrated by arrows 178) may be provided to surface 174.

[0076] As such, various embodiments of the present disclosure provide enhanced cooling about edges of the dilution openings 116. In current designs, per-sector air flow pattern factors vary significantly, which requires a more conservative turbine cooling design. However, and by providing film cooling in accordance with the various embodiments of the present disclosure, the combustor exit temperature profile and uniformity can be controlled using the aforementioned design of air dilution passages 116.

[0077] Since it is difficult to identify the sources of sector-to-sector variation and existing designs are potentially highly sensitive to manufacturing variations embodiments of the present disclosure are particularly useful in providing circumferentially uniform combustor exit temperature.

[0078] The various embodiments of the present disclosure may reduce variation in mixing between dilution jets, as the size and dynamics of the recirculation region could modulate the jet penetration and mixing.

[0079] In addition, the backstop 121 guides flow into and through the dilution passage 116. The ensures that flow in and through each dilution passage 116 is the same, reducing passage-to-passage (and hence sector-to-sector) variation in mixing between dilution jets and combustor head end flow.

[0080] As such, the present disclosure provides an apparatus and method for providing uniformity to the combustor exit temperature profile, which in turn can increase turbine life and performance.

[0081] The use of the terms “a” and “an” and “the” and similar references in the context of description (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or specifically contradicted by context. The modifier “about” used in connection with a quantity is inclusive of the stated value and has the meaning dictated by the context (e.g., it includes the degree of error associated with measurement of the particular quantity). All ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. It should be appreciated that relative positional terms such as “forward,” “aft,” “upper,” “lower,” “above,” “below,” and the like are with reference to the normal operational attitude of the vehicle and should not be considered otherwise limiting.

[0082] Although the different non-limiting embodiments have specific illustrated components, the embodiments of this invention are not limited to those particular combinations. It is possible to use some of the components or features from any of the non-limiting embodiments in combination with features or components from any of the other non-limiting embodiments.

[0083] It should be appreciated that like reference numerals identify corresponding or similar elements throughout the several drawings. It should also be appreciated that although a particular component arrangement is disclosed in the illustrated embodiment, other arrangements will benefit therefrom.

[0084] Although particular step sequences are shown, described, and claimed, it should be understood that steps may be performed in any order, separated or combined unless otherwise indicated and will still benefit from the present disclosure.

[0085] The foregoing description is exemplary rather than defined by the features within. Various non-limiting embodiments are disclosed herein, however, one of ordinary skill in the art would recognize that various modifications and variations in light of the above teachings will fall within the scope of the appended claims. It is therefore to be appreciated that within the scope of the appended claims, the disclosure may be practiced other than as specifically described. For that reason, the appended claims should be studied to determine true scope and content.

What is claimed is:

1. A wall assembly for use in a combustor of a gas turbine engine, the wall assembly comprising:

a support shell;

a liner panel; and

an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage;

a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and

cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

2. The wall assembly as in claim 1, further comprising an internal plenum located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum.

3. The wall assembly as in claim 2, further comprising a plurality of openings located in the surface of the backstop above the support shell, the plurality of openings being in fluid communication with the internal plenum.

4. The wall assembly as in claim 2, further comprising a protrusion that extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion.

5. The wall assembly as in claim 4, wherein the at least one opening is a plurality of individual openings or a single elongated slot.

6. The wall assembly of claim 5, wherein the protrusion has a curvature that matches a curvature of the dilution passage.

7. The wall assembly of claim 6, wherein the plurality of individual openings or the single elongated slot are arranged in an arc.

8. The wall assembly of claim 1, wherein the backstop is arranged to be at least partially located at a trailing edge portion of the dilution passage.

9. The wall assembly as in claim 1, further comprising a plurality of openings located in the surface of the backstop, a surface of the annular peripheral wall defining the dilution passage and a bottom portion of the annular peripheral wall, the plurality of openings being in fluid communication with the cooling airflow paths and the cooling airflow paths are angled with respect to the surface of the backstop and the bottom portion of the annular peripheral wall.

10. The wall assembly as in claim 9, wherein some of the plurality of openings located in the bottom portion of the annular peripheral wall are further from a trailing edge of the dilution passage than other ones of the plurality of openings located in the bottom portion of the annular peripheral wall, the some of the plurality of openings located in the bottom portion of the annular peripheral wall that are further from the trailing edge of the dilution passage correspond to some of the plurality of openings located in the surface of the backstop or the surface of the annular peripheral wall that are further from the bottom portion of the annular peripheral wall than others of the plurality of openings located in the surface of the backstop or the surface of the annular peripheral wall.

11. The wall assembly as in claim 1, further comprising: an internal plenum located in the annular peripheral wall, the cooling airflow paths extending through the internal plenum and at least one opening located in a surface of the annular peripheral wall defining the dilution passage, the at least one opening being in fluid communication with the internal plenum;

a protrusion that extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion.

sion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion.

12. The wall assembly as in claim **11**, wherein the at least one opening located in the trailing edge portion of the protrusion is a plurality of individual openings or a single elongated slot.

13. The wall assembly of claim **12**, wherein the protrusion has a curvature that matches a curvature of the dilution passage.

14. The wall assembly of claim **13**, wherein the plurality of individual openings or the single elongated slot are arranged in an arc.

15. The wall assembly as in claim **1**, further comprising: an internal plenum located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum; and

a plurality of openings located in the surface of the backstop above the support shell, a surface of the annular peripheral wall defining the dilution passage and a bottom portion of the annular peripheral wall, the plurality of openings being in fluid communication with the internal plenum.

16. The wall assembly as in claim **1**, further comprising: an internal plenum located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum;

a protrusion that extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion;

a plurality of openings located in the surface of the backstop above the support shell, and a surface of the annular peripheral wall defining the dilution passage, the plurality of openings and the at least one opening located in a trailing edge portion of the protrusion being in fluid communication with the internal plenum.

17. The wall assembly as in claim **16**, wherein the protrusion completely encircles the dilution passage and a bottom portion of the protrusion has openings in fluid communication with the internal plenum.

18. The wall assembly as in claim **16**, wherein the internal plenum extends completely around an interior of the annular peripheral wall.

19. A gas turbine engine, comprising:

a compressor section;

a turbine section; and

a combustor section, the combustor section including a wall assembly for use in a combustor of the combustor section, the wall assembly comprising:

a support shell;

a liner panel; and

an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage;

a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and

cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

20. A method of guiding airflow into a combustor of a gas turbine engine, the method comprising:

locating a portion of an annular peripheral wall in an opening of a support shell of the combustor, the annular peripheral wall extending from a liner panel secured to the support shell, the annular peripheral wall defining a dilution passage; and

directing airflow into the dilution passage via a backstop extending from a top portion of the annular peripheral wall, the top portion defining a portion of a periphery of the dilution passage; and

directing airflow into cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

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